

Communication-Efficient Semantic Split Federated Learning via VIB-Driven Dynamic Slicing

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Abstract—Split Federated Learning (SFL) offers a compute-efficient and privacy-preserving solution for resource-constrained Internet of Things (IoT) systems by offloading heavy computations to servers. However, SFL requires transmitting high-dimensional intermediate features at every batch, leading to severe communication bottlenecks and vulnerability to wireless channel noise. Nevertheless, existing approaches have not considered wireless channel state and semantics of those intermediate features jointly during SFL training. Accordingly, this letter proposes a novel Channel-Aware Semantic SFL (CA-SSFL) framework. Specifically, the client model employs a dual-masking architecture. First, a distributed variational information bottleneck (VIB) evaluates the semantic priority of each feature dimension and generates a semantic mask. Concurrently, a Signal-to-Noise Ratio (SNR)-based module actively converts the SNR into a dynamic channel mask to define the available transmission budget. By computing the element-wise product of this semantic mask and the channel mask, we introduce an adaptive dynamic slicing mechanism that selectively reduces irrelevant or unreliable feature dimensions under current channel conditions. Extensive experiments on the CIFAR-10, CIFAR-100 dataset under non-IID settings demonstrate that the proposed CA-SSFL drastically reduces communication costs while outperforming existing methods in both accuracy and computational latency under severe noise conditions.

Index Terms—Split Federated Learning, Semantic Communication, Communication Efficiency

I. INTRODUCTION

WITH the explosive proliferation of Internet of Things (IoT) devices, Federated Learning (FL) has emerged as a promising paradigm for edge intelligence, enabling collaborative learning while preserving data privacy. However, existing FL methods require each device to learn the entire global model, significant computational and energy burden on resource-constrained IoT devices [1], [2]. Split Federated Learning (SFL) [3] addresses this limitation by splitting the model between clients and a central server. By offloading significant computational overhead to the server, SFL dramatically reduces the local computational overhead while maintaining parallel collaborative learning.

However, the computational benefits of SFL come at the cost of a severe communication bottleneck. Unlike FL, which periodically exchanges model weights, SFL requires clients to transmit high-dimensional intermediate feature maps, commonly referred to as smashed data, to the server at every training batch [4], [5]. This frequent exchange of massive data streams generates massive communication overhead and latency, especially over bandwidth-limited IoT wireless networks.

Recent studies have mainly focused on improving communication efficiency in SFL through methods such as quantization, sparsification, and pruning [6], [7]. While these approaches have achieved some success in reducing communication traffic, they fundamentally operate by mechanically compressing data

without considering its semantic relevance to the downstream task. For example, magnitude-based pruning or activation sparsification removes low-magnitude activations [8], [9], but this can remove features that might be very important for distinguishing classes, which can significantly reduce inference accuracy, especially at high compression rates. Consequently, conventional approaches fundamentally lack a semantic understanding of the transmitted data. To overcome this limitation, semantic communication has emerged as a practical paradigm shift. Instead of aiming for exact bit-level reconstruction, it focuses on extracting and transmitting only the core semantic information essential for the downstream task [10], [11]. However, integrating these semantic communication principles into SFL frameworks is still in an early stage with limited literature. A few pioneering studies have begun to explore this intersection to address severe communication bottlenecks in distributed edge [12], [13].

In particular, the work of [12] proposed a mobility-aware SFL framework that divides the semantic coder into multiple components to reduce the computational load on vehicles, utilizing transfer learning to efficiently adapt to highly dynamic environments. Building upon this direction, the work of Yu et al. [13] introduced a U-shaped SFL architecture equipped with a task-oriented semantic communication module, demonstrating improved robustness against wireless channel noise through an external network status monitor. Despite these advances, existing frameworks still have structural limitations that restrict their practical use in dynamic IoT systems. First, rather than enabling the neural network to adapt to dynamic channel conditions, these frameworks heavily depend on external network-state monitors and complex optimization algorithms to determine compression rates. Second, training the SFL backbone model and semantic module together under dynamic channel conditions causes instability, so existing methods use two-stage training with a fixed semantic module, which makes real-time end-to-end optimization difficult [14], [15]. Third, their effectiveness has not been sufficiently validated under non-independent and identically distributed (non-IID) data settings, which remains a fundamental challenge in realistic SFL scenarios.

To address these limitations, it is essential to enable neural networks to recognize channel states and adaptively compress smashed data. To the best of our knowledge, this is the first study to integrate active channel-aware semantic communication into the SFL framework, thereby reducing communication overhead tailored to the wireless environment. In this letter, we propose a novel Channel-Aware Semantic Split Federated Learning (CA-SSFL) framework over wireless networks to overcome the aforementioned limitations. The main contributions are summarized as follows. 1) We propose a new dual-masking architecture that integrates a Variational Information Bottleneck (VIB)-driven Semantic Masking Module and a Signal-to-Noise Ratio (SNR)-driven Channel Masking Module. This enables the client model to evaluate the semantic importance of features and the instantaneous wireless channel state, essentially, 2) By integrating these two masks, we introduce an adaptive dynamic slicing mechanism. This mechanism selectively reduces irrelevant or unreliable feature dimensions under current channel conditions, thereby enabling autonomous variable-length transmission without relying on

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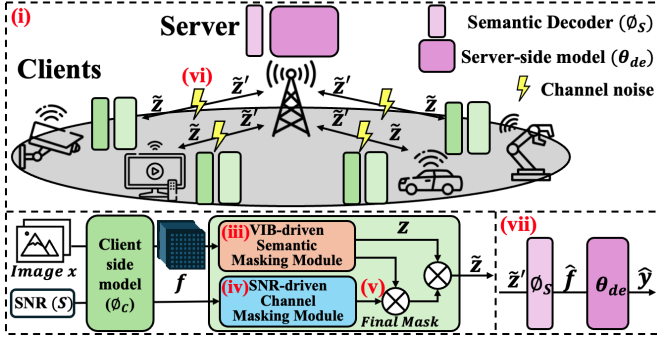


Fig. 1: System model of the proposed scheme.

an external control module. 3) Extensive experiments on the CIFAR-10, CIFAR-100 dataset under non-IID settings demonstrate that the proposed CA-SSFL significantly reduces communication overhead while achieving higher accuracy and lower latency than benchmarks.

II. SYSTEM MODEL

A. Proposed CA-SSFL Architecture

As shown in (i) of Fig. 1, the proposed CA-SSFL system consists of multiple resource-constrained clients (e.g., IoT devices) and a single edge server with abundant computational resources. To concurrently address the dual constraints of limited computational capabilities in IoT devices and insufficient communication bandwidth in wireless networks, we adopt a SFL architecture [3], [4]. This architecture divides the entire global model Θ into two sub-models: the client-side model Θ_C and the server-side model Θ_S . Specifically, the client-side model includes a feature extractor by ϕ_c and a semantic encoder by θ_{en} that compresses the smashed data before transmission (i.e., $\Theta_C = \{\phi_c, \theta_{en}\}$). On the other hand, the server-side model consists of a semantic decoder by θ_{de} to reconstruct the received data, and the subsequent deep neural network layers by ϕ_s to perform the image classification task considered in this work (i.e., $\Theta_S = \{\theta_{de}, \phi_s\}$). Therefore, the total model set of the proposed framework is defined as $\Theta = \{\phi_c, \theta_{en}, \theta_{de}, \phi_s\}$.

B. Overall Process

The data transmission and processing pipeline of the proposed system is formulated as follows. As shown in (ii) of Fig. 1, the client takes an image vector $x \in \mathbb{R}^{C \times W \times H}$ (where C , W , and H represent the channel dimension, image width and height, respectively) and a scalar SNR value s , representing the current wireless channel state, as inputs. The client-side model ϕ_c , acting as a feature extractor, processes the input image x to generate an smashed data f :

$$f = \phi_c(x). \quad (1)$$

Subsequently, the generated smashed data f and the channel state s are provided to the semantic encoder θ_{en} , which compresses them into a complex-valued channel symbols $\tilde{z} \in \mathbb{C}^k$ optimized for the current wireless environment:

$$\tilde{z} = \theta_{en}(f, s). \quad (2)$$

Since IoT devices operate in time-varying wireless environments, a static encoder often fails to extract meaningful semantic features across dynamic channel state. To address this, we designed a semantic encoder θ_{en} that integrally combines a channel-aware mechanism and a probabilistic filtering process via a parallel dual-masking architecture. This coupling allows

the model to dynamically modulate the smashed data based on the current channel state, ensuring better semantic slicing.

Specifically, as shown in (iii) of Fig. 1, the VIB-driven semantic masking module evaluates the informational value of each dimension in f . Rather than mechanically filtering data, this module generates a semantic mask that calculates the task relevance of features and gives priority to essential information. Concurrently, as illustrated in (iv) of Fig. 1, the SNR-driven channel masking module utilizes the current SNR s to generate a channel-aware mask. This module acts as an active gatekeeper, translating physical channel constraints into a transmission budget. Finally, as shown in (v) of Fig. 1, these two masks are fused via element-wise multiplication to form a 'Final Mask'. This mask is then applied to the latent vector z , which is sampled through a reparameterization process, to produce the final smashed data $\tilde{z}$. Through this dual-masking design, the encoder performs dynamic semantic slicing that extracts only the features satisfying both semantic importance and current channel state. Detailed operations of these modules will be further discussed in Section III.

After power normalization, as shown in (vi) of Fig. 1, the semantically encoded vector $\tilde{z} \in \mathbb{C}^k$ is transmitted over a noisy wireless channel [16]. The received signal z' is modeled as:

$$\tilde{z}' = h \cdot \tilde{z} + n, \quad (3)$$

where $n \sim \mathcal{CN}(0, \sigma_n^2 I_k)$ represents the independent and identically distributed (i.i.d.) complex additive white Gaussian noise (AWGN) vector. Here, σ_n^2 denotes the noise variance, and I_k is the $k \times k$ identity matrix corresponding to the dimension of the transmitted symbol $\tilde{z}$. The channel gain h represents the channel fading coefficient; specifically, $h = 1$ for the AWGN channel and $h \sim \mathcal{CN}(0, 1)$ for the Rayleigh fading channel.

Finally, as shown in (vii) of Fig. 1, the semantic decoder θ_{de} located at the server side decodes the received corrupted symbol $\tilde{z}'$ to transform it into task-relevant smashed data $\hat{f}$ tailored for the downstream task.

$$\hat{f} = \theta_{de}(\tilde{z}'), \quad (4)$$

In this process, zero padding is used to variably adjust the dimension of the transmitted data to match the decoder's fixed input size. The decoded smashed data then passes through the server-side model ϕ_s , consisting of deep layers, to produce the final prediction $\hat{y}$:

$$\hat{y} = \phi_s(\hat{f}). \quad (5)$$

By enabling the client's θ_{en} to autonomously compress and transmit the compressed smashed data ($\tilde{z}$) according to the SNR state, the communication bottleneck in bandwidth-constrained environments is effectively overcome. To achieve this, the entire global model Θ is trained by a loss function that balances the trade-off between the downstream task accuracy and the communication overhead. This loss function guides the VIB-driven Semantic Masking module to filter out semantically irrelevant information while ensuring that the SNR-driven Channel Masking module generates a dynamic mask appropriate for the current channel state. The detailed loss function design is presented in the following subsection.

C. Loss Function Design

The proposed CA-SSFL is designed to ensure high accuracy in downstream tasks under a given channel state s . Additionally, it reduces the size of the smashed data transmitted by clients to enhance communication efficiency. To achieve this, we adopt the Information Bottleneck (IB) principle [17]. We formulate the loss function as the total loss $\mathcal{L}_{Total}$, which consists of the task-specific loss $\mathcal{L}_{Task}$ and the information

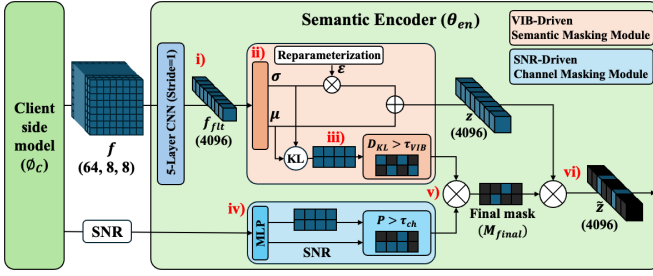


Fig. 2: Proposed semantic encoder.

compression loss $\mathcal{L}_{\text{Compression}}$, balanced by a hyperparameter β . β controls the balance between the model's task performance and the communication data compression rate. Depending on β , the model reduces the information of features irrelevant to the task to zero. This loss function allows the entire model to be trained in an end-to-end manner, as follows:

$$\min_{\phi_c, \theta_{en}, \theta_{de}, \phi_s} \mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Task}} + \beta \cdot \mathcal{L}_{\text{Compression}}. \quad (6)$$

To apply this theoretical loss function to actual neural network training, we utilize the variational upper bound proposed in the VIB framework [18] and reformulate the final loss function as:

$$\mathcal{L}_{\text{Total}} = \mathbb{E}_{z \sim p_{\theta_{en}}(z|\phi_c(x))} [-\log q_{\theta_{de}, \phi_s}(y|\tilde{z}')] + \beta \cdot D_{KL}(p_{\theta_{en}}(z|\phi_c(x)) \parallel r(z)). \quad (7)$$

In (7), the first term denotes the task loss, measured by the cross-entropy. It evaluates how accurately the server-side model $q_{\theta_{de}, \phi_s}(y|\tilde{z}')$ predicts the ground-truth label y , based on $\tilde{z}'$, which is the channel-distorted version of the dynamically sliced latent vector. The second term represents compression loss. Here, β represents the penalty weight governing the information bottleneck. This penalty constrains uninformative dimensions to approximate the standard normal prior distribution $r(z) \sim \mathcal{N}(0, I_D)$ (where D denotes the total number of latent dimensions). Here, k denotes the dynamically determined dimension of the semantic vector after adaptive slicing. I_k denotes the $k \times k$ identity matrix corresponding to this active latent space, and the standard normal prior $r(z)$ represents an uninformative state. By reducing this divergence, the encoder is penalized for transmitting excess information. This enables the encoder to focus the most important semantic features into fewer dimensions.

III. PROPOSED METHOD

A. VIB-Driven Semantic Masking Module

As shown in (i) of Fig. 2, the flattened intermediate features $f_{fit} \in \mathbb{R}^k$ extracted from the client-side model are directly processed by the VIB-Driven Semantic Masking Module. Unlike conventional SFL frameworks that encode inputs as fixed deterministic points, this module maps the features into a probabilistic latent space characterized by the mean μ and variance σ^2 . It injects random noise during training to enhance semantic robustness against potential channel distortions. Furthermore, it evaluates the informational value of each feature dimension based on its relevance to the downstream task through dimension-wise KL divergence computed from latent statistics modulated by the current channel condition. Finally, this module identifies task-irrelevant dimensions to generate a semantic mask, which will later be fused with the channel masking module to guide the dynamic slicing process.

1) *Stochastic Encoding*: As shown in (ii) of Fig. 2, the semantic encoder θ_{en} estimates the mean μ and variance σ^2 of the latent distribution from f_{fit} . To enable backpropagation through the stochastic sampling process, we apply the reparameterization trick [19]. The latent vector z is sampled as:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (8)$$

where ϵ is an auxiliary noise variable sampled from a standard normal distribution. This process internalizes the uncertainty of the wireless channel into the latent space of the deep learning model.

2) *Semantic Masking via VIB*: As shown in (iii) of Fig. 2, we utilize the channel-wise KL divergence as a metric for information significance [19]. The KL divergence for the k -th dimension between the posterior $p(z_k|x, s)$ and the prior $r(z_k)$ is calculated as:

$$D_{KL}^{(k)} = -\frac{1}{2} \left(1 + \log(\sigma_k^2) - \mu_k^2 - \sigma_k^2 \right). \quad (9)$$

During the forward pass, dimensions with a $D_{KL}^{(k)}$ value strictly greater than a predefined semantic threshold τ_{VIB} are considered highly informative. Based on this evaluation, the module generates a binary semantic mask $M_{vib} \in \{0, 1\}^K$:

$$M_{vib}^{(k)} = \mathbb{I}(D_{KL}^{(k)} > \tau_{VIB}), \quad (10)$$

where $\mathbb{I}(\cdot)$ is the indicator function. This semantic mask M_{vib} is passed to the subsequent fusion stage to be jointly considered with the instantaneous wireless channel constraints.

B. SNR-Driven Channel Masking Module

In wireless networks, the available communication capacity fluctuates severely due to dynamic channel conditions. To address this physical constraint, we introduce the SNR-Driven Channel Masking Module, which acts as an active physical gatekeeper to dynamically allocate the transmission budget.

As shown in (iv) of Fig. 2, this module takes the normalized SNR $s_{norm} \in [0, 1]$ as input. A multilayer perceptron (MLP) processes s_{norm} to generate a channel-specific probability vector $P \in [0, 1]^K$, where K is the total dimension of the latent representation. Each element of P represents the module's confidence that the corresponding feature dimension can be successfully transmitted under the current channel conditions. To actively control the transmission overhead, we introduce a dynamic threshold τ_{ch} . Rather than relying on a fixed cutoff, τ_{ch} is adaptively scaled within a predefined operational range $[\tau_{min}, \tau_{max}]$ parameterized by the physical channel limits:

$$\tau_{ch} = \tau_{max} - s_{norm} \times (\tau_{max} - \tau_{min}). \quad (11)$$

In (11), under low-SNR conditions (e.g., $s_{norm} \rightarrow 0$), the threshold approaches τ_{max} . This enforces strong compression, preventing bandwidth waste on features that are likely to be distorted by severe channel noise. In contrast, under high SNR conditions (e.g., $s_{norm} \rightarrow 1$), the threshold decreases toward τ_{min} . This allows for more efficient use of bandwidth by transmitting more semantic features over low-noise channels, thereby improving task accuracy. Finally, the binary channel mask $M_{ch} \in \{0, 1\}^K$ is obtained by applying a gating mechanism that compares the probability vector P against the dynamic threshold τ_{ch} :

$$M_{ch}^{(k)} = \mathbb{I}(P^{(k)} > \tau_{ch}), \quad (12)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Operating in parallel with the VIB-driven Semantic Masking module, this SNR-driven Channel Masking module translates physical channel limitations into a binary format, preparing to filter out features that are unviable for transmission.

TABLE I: Performance analysis of CA-SSFL on CIFAR-10 AWGN (**B**: Best, **U**: Second)

β	τ_{VIB}	Acc. (%)	Comm. (GB)	τ_{max}	τ_{min}	Acc. (%)	Comm. (GB)
0.100	1.5	20.34	8.22	0.7	0.2	41.71	14.03
	1.0	39.41	12.63		0.3	40.70	14.05
	0.5	40.20	15.51		0.4	41.39	13.41
0.050	1.5	16.57	8.19	0.8	0.2	40.44	13.23
	1.0	41.39	13.41		0.3	39.80	12.85
	0.5	39.82	17.39		0.4	<u>41.53</u>	12.52
0.010	1.5	32.38	8.66	0.9	0.2	40.08	12.64
	1.0	40.46	15.87		0.3	40.83	12.05
	0.5	37.66	24.05		0.4	41.27	11.55
0.005	1.5	32.69	9.27	* Fixed Settings - Left: $\tau_{max}=0.7$, $\tau_{min}=0.4$ - Right: $\beta=0.05$, $\tau_{VIB}=1.0$			
	1.0	40.83	17.12				
	0.5	40.69	31.12				
0.001	1.5	39.04	9.63				
	1.0	40.98	20.59				
	0.5	<u>38.20</u>	39.41				

C. Dual-Masking and Dynamic Slicing

To integrally account for both the semantic importance of the features and the current channel capacity, we propose a dual-masking mechanism. As shown in (v) of Fig. 2, the final transmission mask $M_{final} \in \{0, 1\}^K$ is derived by the element-wise multiplication of the semantic mask and the channel mask:

$$M_{final} = M_{vib} \odot M_{ch}. \quad (13)$$

This formulation ensures that a feature dimension is preserved only if it is both relevant to the downstream task and viable for transmission under the instantaneous channel conditions. Subsequently, the sampled latent vector z is masked by this combined decision: $z_{masked} = z \odot M_{final}$. However, transmitting z_{masked} directly does not reduce the communication cost, as the zero-valued elements still occupy the bandwidth. Therefore, we introduce a dynamic slicing operation. The client extracts only the non-zero elements of z_{masked} along with their corresponding position indices. As shown in (vi) of Fig. 2, $\tilde{z}$ is dynamically sliced, reducing the physical communication overhead. By allocating a zero-initialized tensor and deploying the received values to their exact original positions using the transmitted indices, the spatial structure of the semantic features is restored before they are fed into the decoder.

To address the instability and gradient vanishing issues caused by the extreme sparsity of dynamic slicing, we employ sparsity-aware gradient compensation. Specifically, the back-propagated gradients from the server are dynamically scaled by the inverse of the active feature ratio ($1/\max(p, 0.01)$). This scaling mechanism, inspired by the principle of inverted dropout [20], ensures that the expected magnitude of the learning signals remains consistent regardless of the fluctuating transmission budget.

IV. PERFORMANCE EVALUATION

A. Experimental Setup

We evaluated the proposed method using the CIFAR-10 dataset in Additive White Gaussian Noise (AWGN) and Rayleigh fading channels. To simulate a non-IID environment, each client was assigned a local dataset of 3,000 non-overlapping training images. To set up the non-IID distribution, we use the Dirichlet distribution $\text{Dir}(\alpha)$ to partition the CIFAR-10, CIFAR-100 dataset among clients. We set the concentration parameter to $\alpha = 0.5$.

We employ a split ResNet-18 architecture, where the client processes the first layer and the server handles the remaining layers 2, 3, and 4. For semantic communication, the widely

TABLE II: Comparison of Accuracy and Communication Cost under AWGN and Rayleigh Channels

Method	AWGN Channel		Rayleigh Channel	
	Acc. (%)	Comm. (GB)	Acc. (%)	Comm. (GB)
SFL	51.04	150.07	47.43	150.07
SC-USFL	36.69	20.68	38.64	20.68
CA-SSFL (Ours)	41.71	14.03	40.41	14.59
w/o Semantic Masking	37.66	45.43	38.67	18.12
w/o Channel Masking	39.34	17.63	38.19	17.90

TABLE III: Comparison of Computational Overhead and Model Size

Algorithm	MFLOPs			Model Size (MB)		
	Client	Server	Total	Client	Server	Total
SFL	19.76	50.34	70.10	0.57	42.05	42.62
SC-USFL	23.26	53.84	77.10	1.95	43.43	45.38
CA-SSFL (Ours)	35.16	78.20	113.36	2.03	42.89	44.92

validated DeepJSCC architecture [16] serves as the backbone. The following baselines were established for comparison: SFL [3] transmits raw smashed data without semantic compression. SC-USFL [13] is a state-of-the-art semantic SFL approach utilizing a pre-trained semantic module and an external Network Status Monitor (NSM). We train all models over a continuous SNR range of $[-5, 15]$ dB to evaluate their adaptability to dynamic wireless environments. All results were calculated by averaging 4 random seeds.

B. Performance Comparisons

1) *Hyperparameter Sensitivity Analysis*: Table I shows the trade-off between classification accuracy and communication overhead of the proposed CA-SSFL, analyzing the effects of the VIB parameters (β, τ_{VIB}) and the dynamic threshold bounds (τ_{max}, τ_{min}). Regarding the VIB module (left), increasing the semantic threshold τ_{VIB} strictly reduces the communication overhead. Interestingly, lower τ_{VIB} values (e.g., 0.5), which permit more data transmission, do not guarantee better accuracy. In fact, setting the threshold near $\tau_{VIB} = 1.0$ or 1.5 can reduce communication overhead while maintaining classification accuracy. This indicates that moderately aggressive slicing effectively filters out task-irrelevant noise and prevents overfitting in non-IID settings, whereas transmitting redundant features degrades global convergence. Furthermore, the right-hand side of the table illustrates the effect of channel-aware dynamic thresholds. The optimal balance was achieved by fixing $\beta = 0.05$ and $\tau_{VIB} = 1.0$, and setting $\tau_{max} = 0.7$ and $\tau_{min} = 0.2$. We performed the same experiment on the Rayleigh channel, fixing $\beta = 0.1$ and $\tau_{VIB} = 1.0$, and applying $\tau_{max} = 0.7$ and $\tau_{min} = 0.4$. This confirms that adaptively constraining the transmission budget based on instantaneous SNR significantly balances the rate-distortion trade-off.

2) *Performance Comparison with Benchmarks*: Table II shows the proposed CA-SSFL achieves communication efficiency and robustness. On an AWGN channel, standard SFL achieves higher accuracy but transmits 150.07 GB of data. The proposed CA-SSFL maintains comparable accuracy while reducing communication overhead to 14.03 GB. As illustrated in (a) of Fig. 3, the proposed CA-SSFL maintains convergence, overcoming performance degradation under non-IID and low-SNR conditions. As shown in (b) of Fig. 3, the test accuracy is shown across various SNR levels. SFL becomes increasingly vulnerable at low SNR, as high-dimensional raw feature transmission is more easily corrupted by severe channel

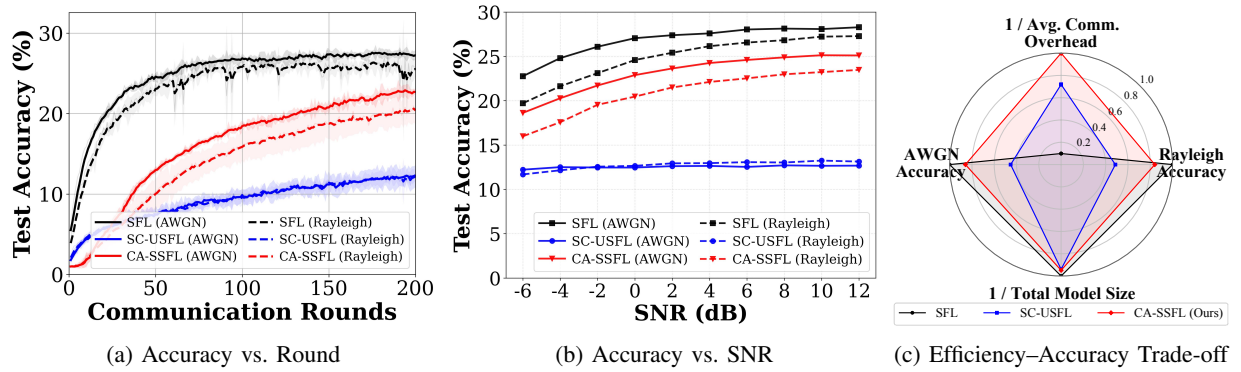


Fig. 3: Performance comparison on CIFAR-100 dataset under Non-IID conditions: (a) test accuracy vs. communication rounds, and (b) test accuracy vs. SNR, and (c) Efficiency–Accuracy Trade-off Radar

noise. Since SC-USFL selects one of four fixed compression ratios based on the channel conditions, it maintains stable performance but suffers from significantly reduced accuracy. In contrast, CA-SSFL demonstrates graceful degradation and maintains superior accuracy over SC-USFL even under low-SNR conditions. This proves that the proposed channel-aware dynamic slicing effectively protects essential semantic information by filtering out noise-sensitive dimensions. Furthermore, as shown in (c), the proposed CA-SSFL achieves a better trade-off between model size, communication overhead, and accuracy.

3) *Ablation Study*: Finally, the ablation analysis results presented at the bottom of Table II highlight the necessity of the proposed dual-masking architecture for both channel types. Removing the VIB-based Semantic Masking module degrades both accuracy and communication efficiency, reducing the accuracy from 41.71% to 37.66% in AWGN and from 40.41% to 38.67% in Rayleigh, while increasing the communication overhead from 14.03 GB to 45.43 GB and from 14.59 GB to 18.12 GB, respectively. The larger communication increase in AWGN can be interpreted together with its channel-specific representative setting, which uses a smaller bottleneck weight ($\beta = 0.05$) than Rayleigh ($\beta = 0.1$), making the semantic mask more critical for reducing unnecessary transmissions. Likewise, removing the SNR-based channel masking module slightly increases communication overhead and reduces accuracy, demonstrating that channel-aware masking is also necessary for an effective accuracy-communication trade-off.

4) *Computational Overhead Comparison*: Table III compares the computational load (MFLOPs) and model size (MB) of various models to evaluate the feasibility of CA-SSFL in resource-constrained IoT environments. Compared to standard SFL, CA-SSFL introduces an additional overhead of 15.40 MFLOPs and 1.46 MB on the client side to operate the dual-masking semantic encoder, which falls well within the processing capabilities of modern IoT devices.

V. CONCLUSION

In this letter, we have proposed a novel CA-SSFL framework that effectively balances the trade-off between communication efficiency and task performance via channel-aware dynamic slicing. The proposed CA-SSFL can be particularly beneficial in resource-constrained IoT scenarios, where low communication overhead, bandwidth efficiency, and robustness against dynamic wireless channels are critical.

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